Nanoscale Device Engineering (EC 502)

Dr. Manish Kumar Rai
Assistant Professor
ECE Department
IIIT Surat

Syllabus

Unit – I L =3, T = 0, P = 0, C = 3

- Introduction to Nanoscience and Nanotechnologies

Unit – 2

- Fabrication Technology & SPICE Device Models

  (VLSI Processes: Introduction, Twin-Well CMOS Process, Integrated Devices, MOSFETs, Resistors, Capacitors, pn Junction Diodes, BiCMOS Process, Lateral pnp Transistor, p-Base and Pinched-Base, Resistors, SiGe BiCMOS Process, VLSI Layout, Beyond 20nm Technology Implications of Technology Scaling: Issues in Deep-Submicron Design: Silicon Area, Scaling Implications, Velocity Saturation, Subthreshold Conduction, PVT Variations, Wiring: The Interconnect, SPICE Device Models: The Diode Model, The Zener Diode Model, MOSFET Models, The BJT Model)

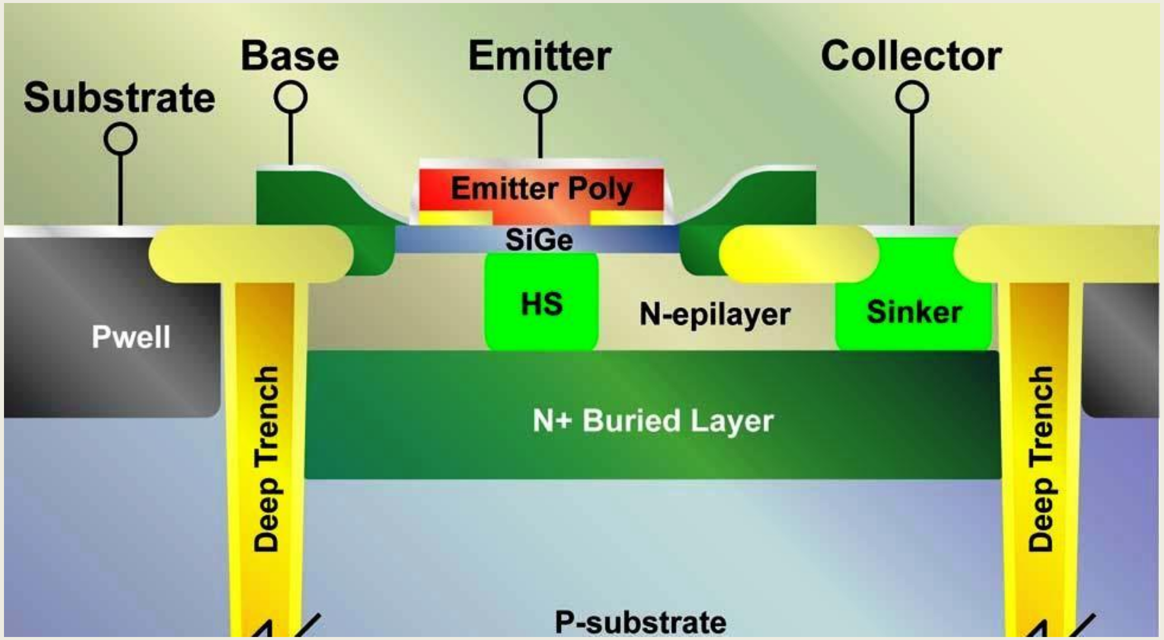
Unit – 3

- Applications of Nanoscience and Nanotechnologies

  (Information and Communication Technologies: Integrated Circuits, Data Storage, Photonics, Displays, Information storage devices, Wireless sensing and communication)

SiGe BiCMOS Process

- SiGe BiCMOS technology integrates the high speed of Silicon-Germanium (SiGe) bipolar transistors with the power efficient CMOS technology on a single chip.
- The addition of germanium to the bipolar transistor's base improves performance, enabling high-frequency applications with high integration and yields.



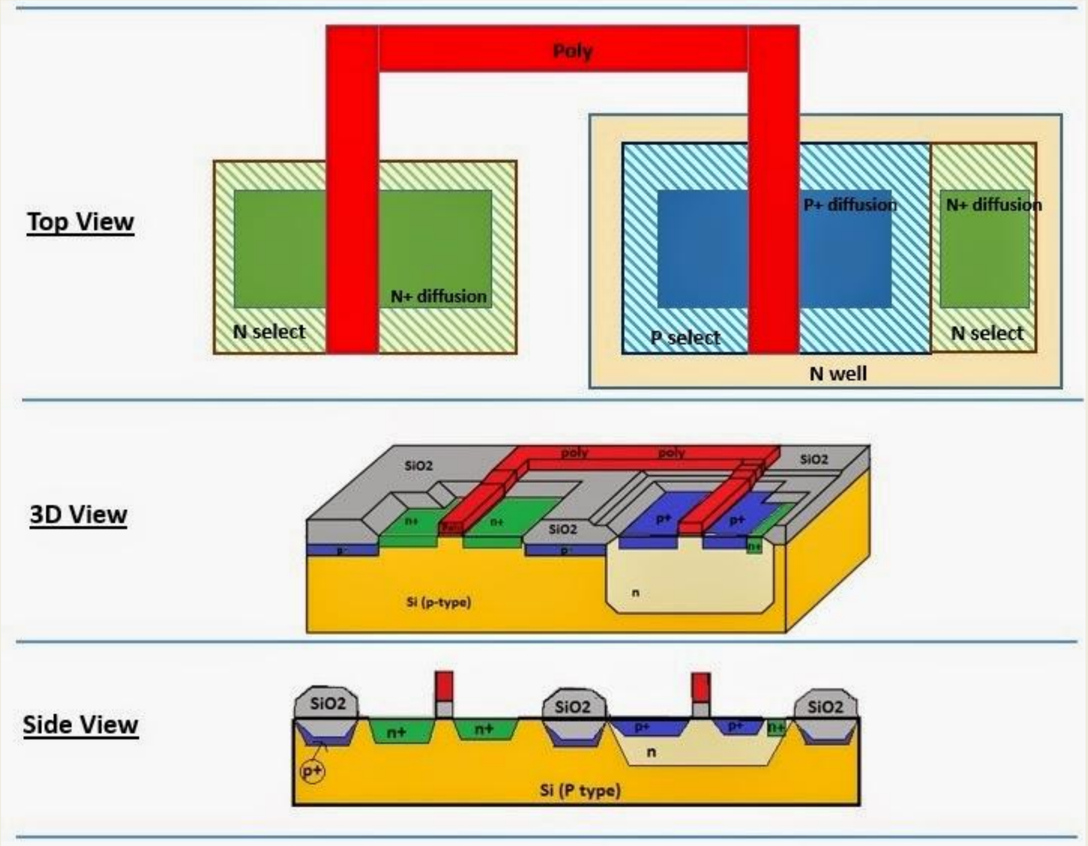
SiGe BiCMOS Process

Advantages of SiGe BiCMOS

1. Bipolar transistors provide high speed which is crucial for high-frequency analog circuits.
2. CMOS technology is excellent for building low-power logic gates, contributing to the overall energy efficiency of the integrated circuit.
3. The technology supports a range of applications, from high-frequency wireless transceivers and high-speed wireline circuits to other radio frequency (RF) and analog applications.
4. SiGe BiCMOS has evolved significantly, achieving performance milestones such as transistors with cut-off frequencies (f_T) of over 200 GHz.

VLSI Layout

- In VLSI, layout is the physical design process of creating the geometric pattern of an electronic circuit on a silicon chip, specifying the precise placement and interconnection of components like MOSFETs, wires and contacts. Layout is the top view of a chip.

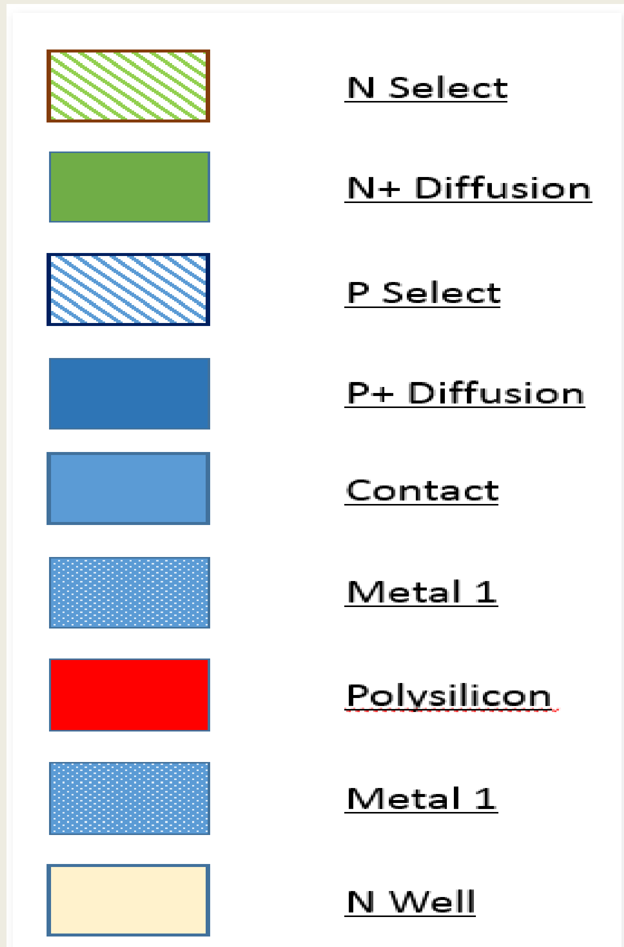


VLSI Layout design

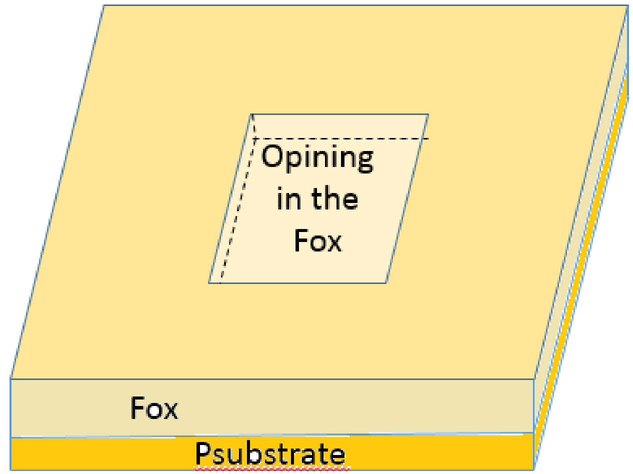
- Design rules are set of geometrical specifications used to design the layout of a VLSI circuit.
- Layout is an interface between circuit designer and fabrication engineer.
- Design rules allow translation of circuit into actual geometry in silicon wafer.
- These rules usually specify the minimum allowable line widths, minimum feature dimension and minimum allowable separation.
- **Objective**
- To achieve a high overall yield and reliability while using the smallest possible silicon area.

VLSI Layout design

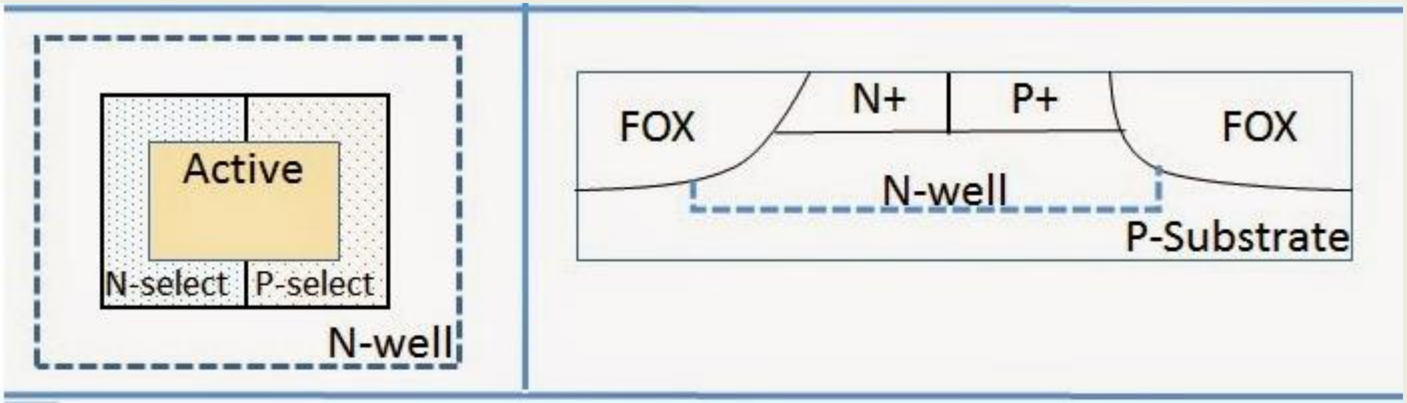
- N-select or P-select layers indicate where to implant n-type or p-type atoms respectively.
- Active layer in a layout defines opening in SiO_2 covering.



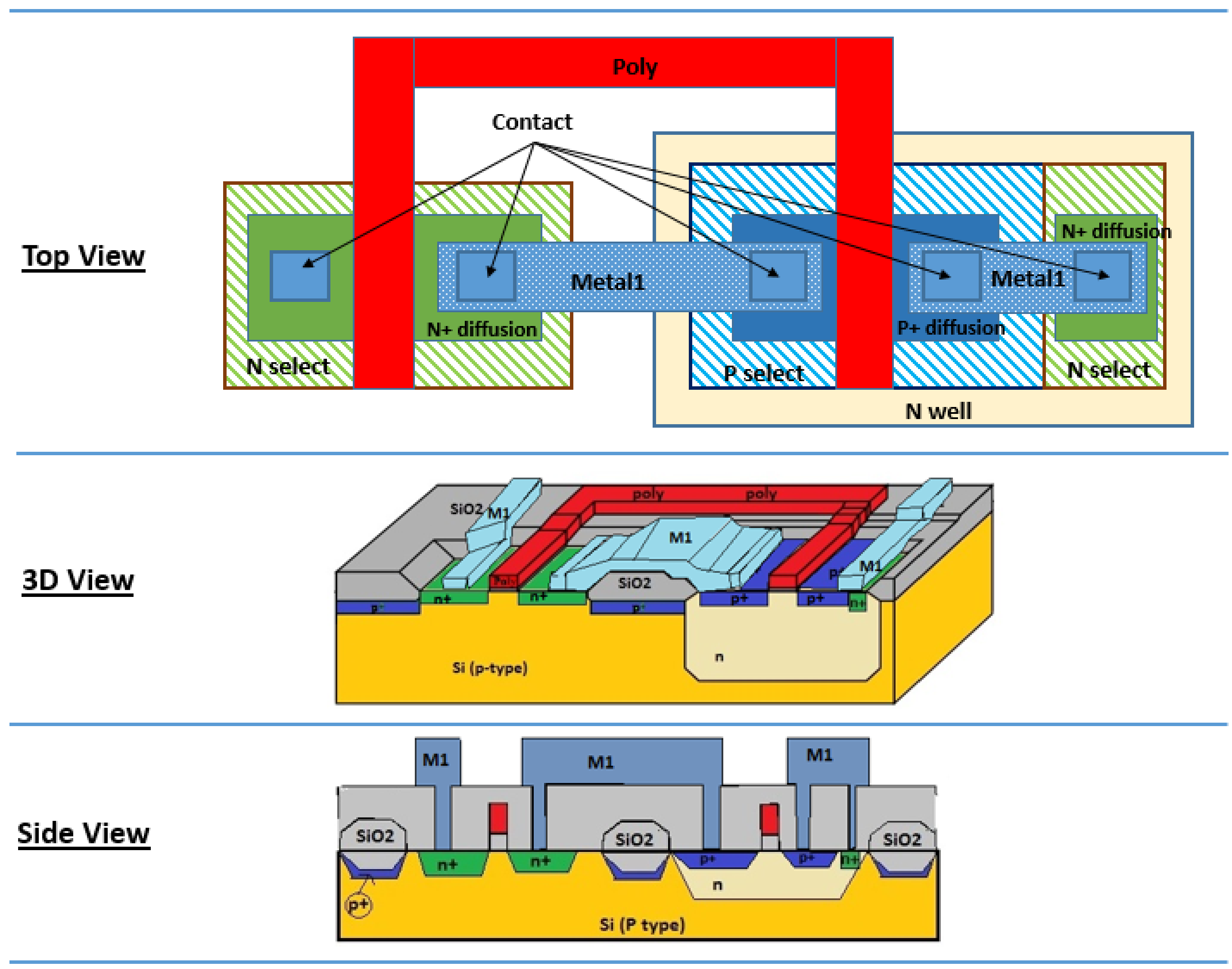
VLSI Layout design



"Active layer" Specifies where to Open Holes in the Field Oxide (FOX)



VLSI Layout design



VLSI Layout design rules

➢ In VLSI two design rules are there:

1. Lambda (λ) based rules

2. Micron (μ) based rules

- λ rules are a set of scalable design rules used for integrated circuit layout, while μ rules (also known as micron-based rules) use absolute micrometer measurements.
- lambda (λ) is a scalable, technology-dependent parameter that represents a fundamental unit of measurement for all dimensions and spacing in an integrated circuit layout.
- **Lambda (λ)** represents **half of the minimum feature size** (or sometimes half of the minimum channel length **λ=L/2**) in a given technology.
- Instead of specifying absolute dimensions (like 0.25 μm or 50 nm), we describe layouts using **multiples of λ**.
- This allows the same layout to be **scaled** easily when moving from one technology node to another.

VLSI Layout design rules

1. **Lambda (λ) based rules**

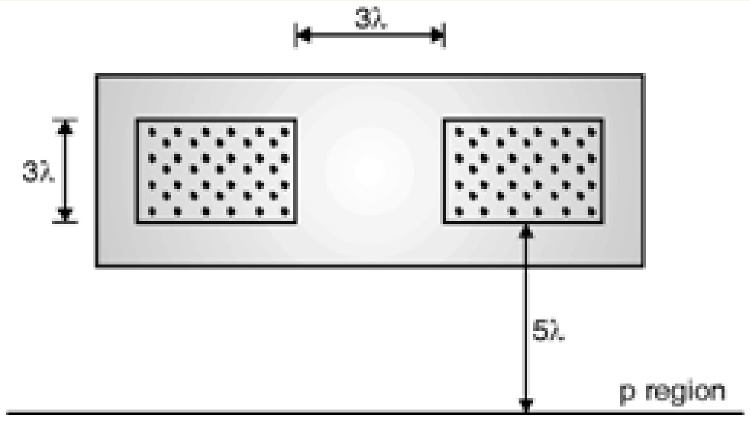
- These rules were first proposed by **Mead and Conway** in 1980.

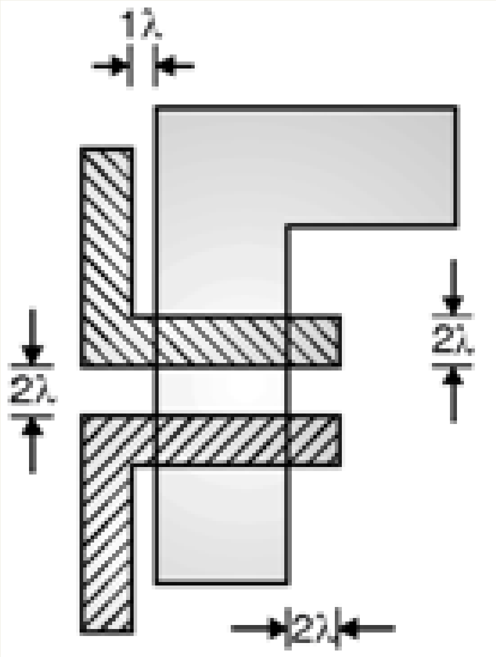
Parameter	Rule (in λ)
Minimum polysilicon width	2λ
Minimum metal width	3λ
Minimum spacing between poly lines	2λ
Minimum spacing between metal lines	3λ
Minimum diffusion width	3λ
Poly to diffusion overlap (for gate)	2λ
Contact size	2λ × 2λ
Contact to diffusion or poly edge	1λ
Metal overlap of contact	1λ

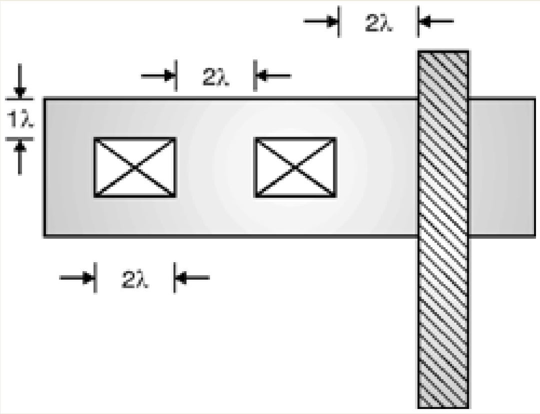
VLSI Layout design rules

e.g. For a simple **nMOS transistor layout** (using λ rules):

- Diffusion width: 3λ
- Gate (poly) width: 2λ
- Poly over diffusion: 2λ overlap
- Contact: 2λ × 2λ
- Metal line width: 3λ







VLSI Layout design rules

2. Micron (μ) based rules

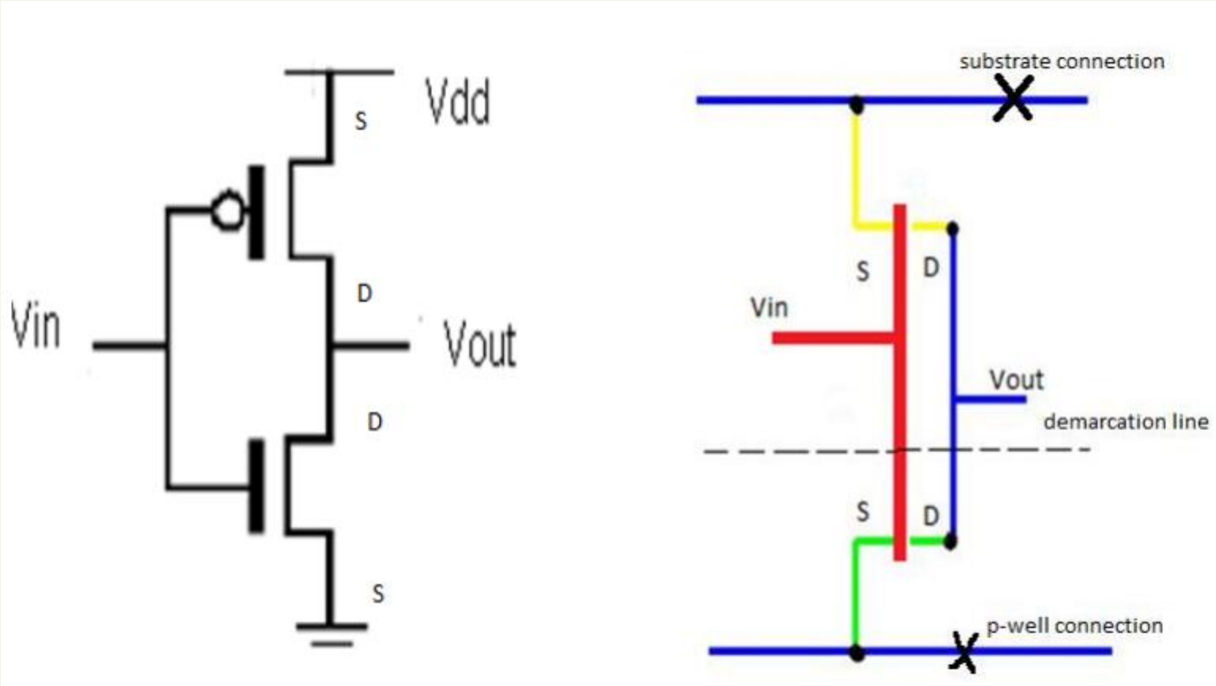
- Unlike λ-based (scalable) rules, these μ-based rules use **absolute dimensions in micrometers** to define layout constraints such as:

TABLE 3.1 Micron design rules for 65 nm process

Layer	Rule	Description	65 nm Rule (μm)
Well	1.1	Width	0.5
	1.2	Spacing to well at different potential	0.7
	1.3	Spacing to well at same potential	0.7
Active (diffusion)	2.1	Width	0.10
	2.2	Spacing to active	0.12
	2.3	Source/drain surround by well	0.15
	2.4	Substrate/well contact surround by well	0.15
	2.5	Spacing to active of opposite type	0.25
Poly	3.1	Width	0.065
	3.2	Spacing to poly over field oxide	0.10
	3.2a	Spacing to poly over active	0.10
	3.3	Gate extension beyond active	0.10
	3.4	Active extension beyond poly	0.10
	3.5	Spacing of poly to active	0.07

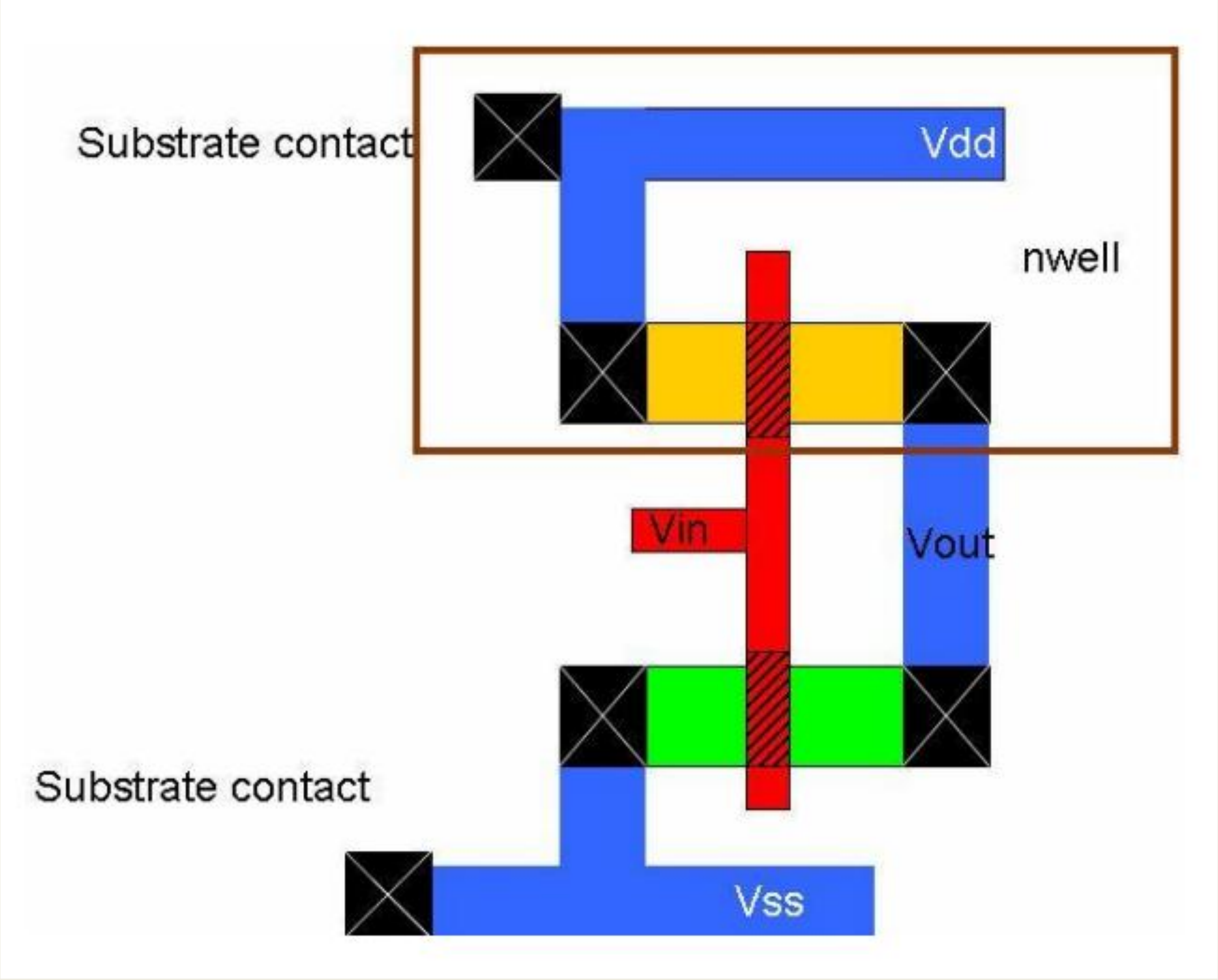
VLSI Layout stick diagram

- Stick diagram is a high-level, cartoon-like schematic representing a circuit's layout, using colored or monochrome lines (sticks).
- It shows the relative placement and electrical connections between different layers like polysilicon, diffusion, and metal, thereby serving as an intermediate step between a circuit's symbolic representation and its detailed physical layout.



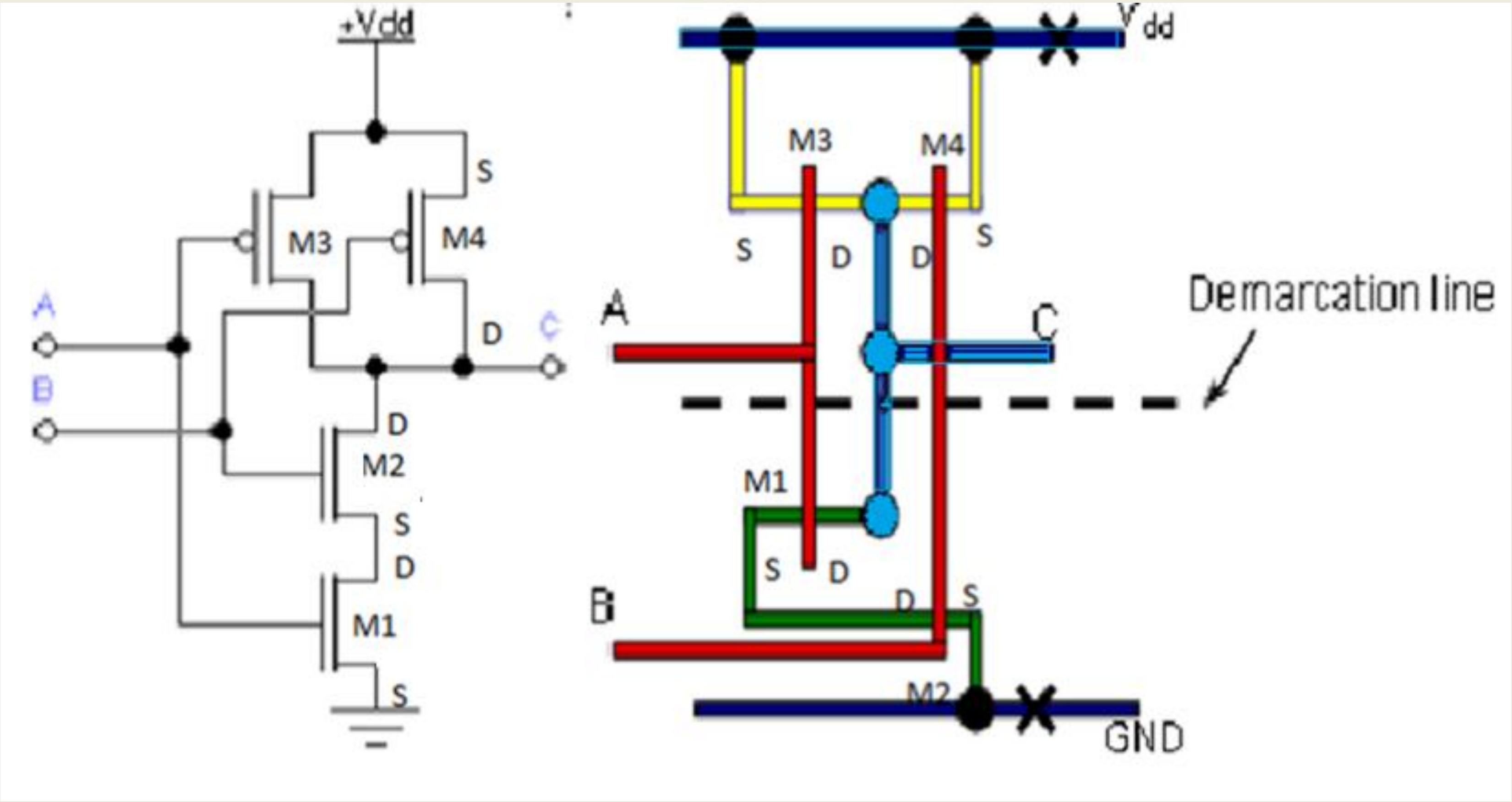
VLSI Layout stick diagram

- Inverter layout



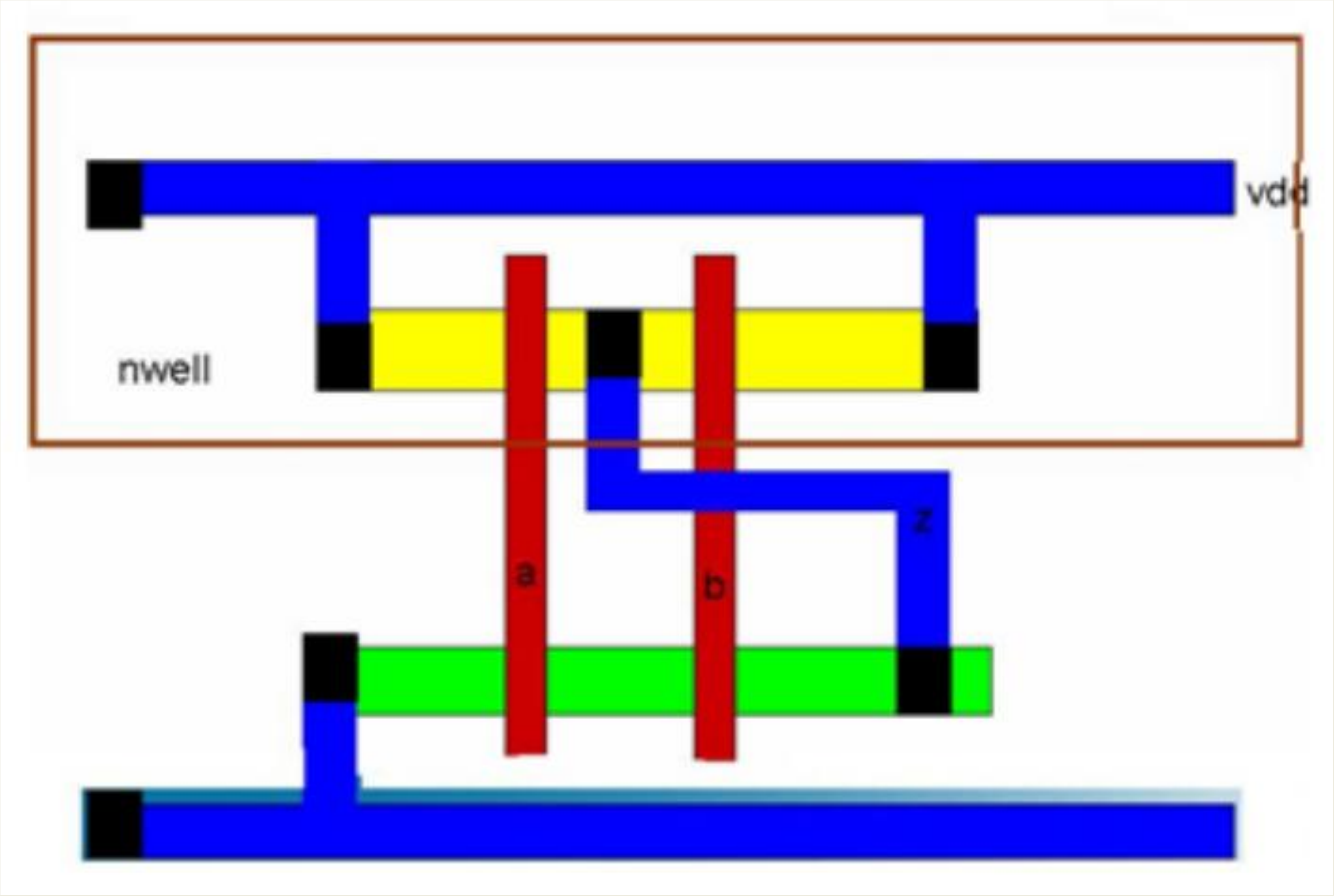
VLSI Layout stick diagram

- NAND



VLSI Layout stick diagram

- NAND



Wiring: The Interconnect

- Fabrication process starts with the wafer. Further oxidation, Lithography and diffusion are used to fabricate the devices.

- All the doping, deposition and etching processes used to fabricate the transistors are called the “front-end process” steps and by analogy, the interconnect layers on top of the devices are called the “back-end” steps.

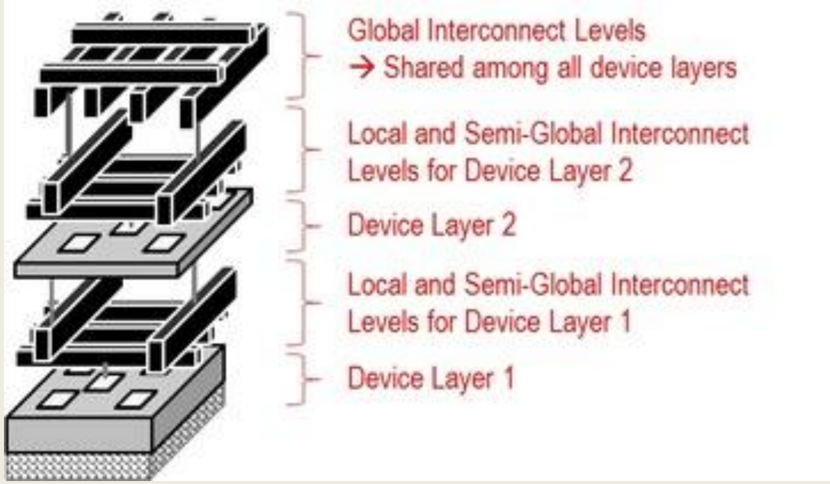
- Interconnects can either be local or global depending upon the length of the wire.

- In general, local interconnects are the first, or lowest, level of interconnects. They usually connect gates, sources and drains in MOS technology and connect nearby transistors to realize circuit functions.

- Local interconnects are short, high-resistance connections that link components within a small area, such as transistors.

Wiring: The Interconnect

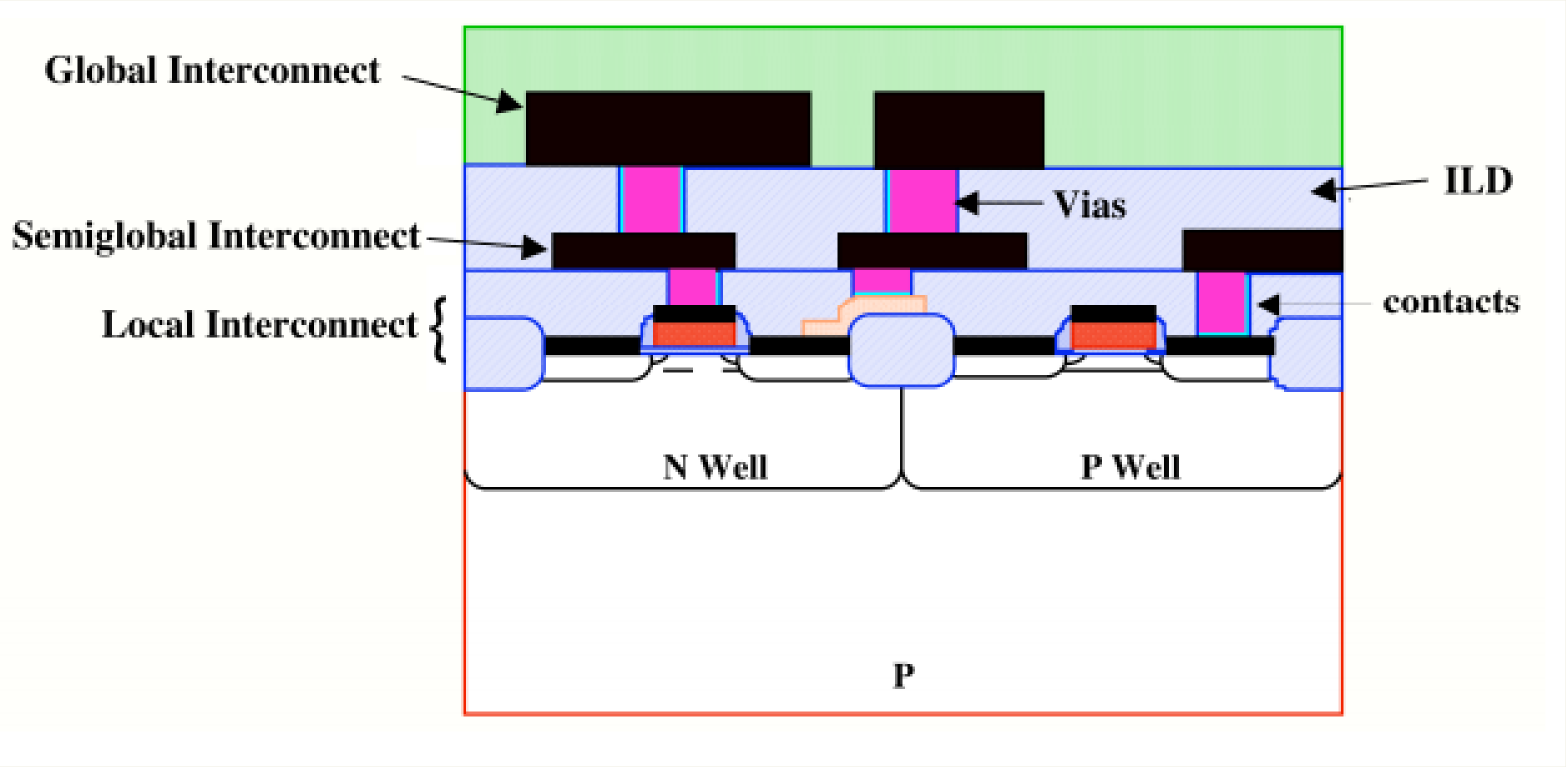
- Global interconnects are long, low-resistance connections that span across the chip to connect different functional blocks.



- Local interconnects can tolerate higher metal resistance than global interconnects since they are not very long.
- In today’s technologies, local interconnects are made from Cu but the metal lines are smaller in cross-section because metal resistance is not as important as it is in longer-distance wires.
- Global interconnects are also mostly made of Cu today. They often cover large distances between different circuit blocks and different parts of the chip.

Wiring: The Interconnect

- Connections between two levels of interconnects are usually called vias.
- They are made through openings in the different levels of the inter-metal dielectrics (IMD), which separate interconnect levels from each other.



Wiring: The Interconnect

- Much of the innovation in back-end materials and process technology that has occurred over the last 50 years, has been aimed at 3 issues:

1. Increasing the number of interconnect levels so that increasing numbers of active components can be successfully wired into complex circuits and systems.

2. Reducing the resistance R of the metal wires.

3. Decreasing the parasitic capacitance C that exists between wires in a complex back-end structure.

Wiring: The Interconnect

IIIT SURAT

- Difference between local and global interconnect:

Feature	Local Interconnect	Global Interconnect
Length	Short, connect nearby components.	Long, connect different functional blocks or across the entire chip.
Purpose	Connect gates, sources, drains, etc., within a small area.	Connect large, separated parts of the circuit, like power or clock lines.
Thickness and Spacing	Thin and closely packed.	Thick and widely spaced.
Material	Can use materials with higher resistivity, such as polysilicon or tungsten, because the distance is short.	Use low-resistivity metals like aluminum or copper to minimize signal delay over long distances.
Scaling	Length generally scales down with the technology node.	Length is a fixed fraction of the die size and does not scale down in the same way.

Wiring: The Interconnect

Properties of interconnect materials:

Material	Thin-film resistivity (μWcm)	Melting point (˚C)
Al	2.7-3.0	660
Cu	1.7-2.0	1084
Ag	≈ 1.6	961
Au	≈ 2.2	1064
W	8-15	3410
Ti	40-70	1670
PtSi	28-35	1229
TiSi2	13-16	1540
WSi2	30-70	2165
CoSi2	15-20	1326
NiSi	14-20	992
TiN	50-150	~2950
Ti0.3W0.7	75-200	~2200
Polysilicon (heavily doped)	450-1000	1417

SPICE Device Models

- A device SPICE (**Simulation Program with Integrated Circuit Emphasis)** model is a mathematical description of an electronic component's behavior used by SPICE circuit simulators to predict its performance.

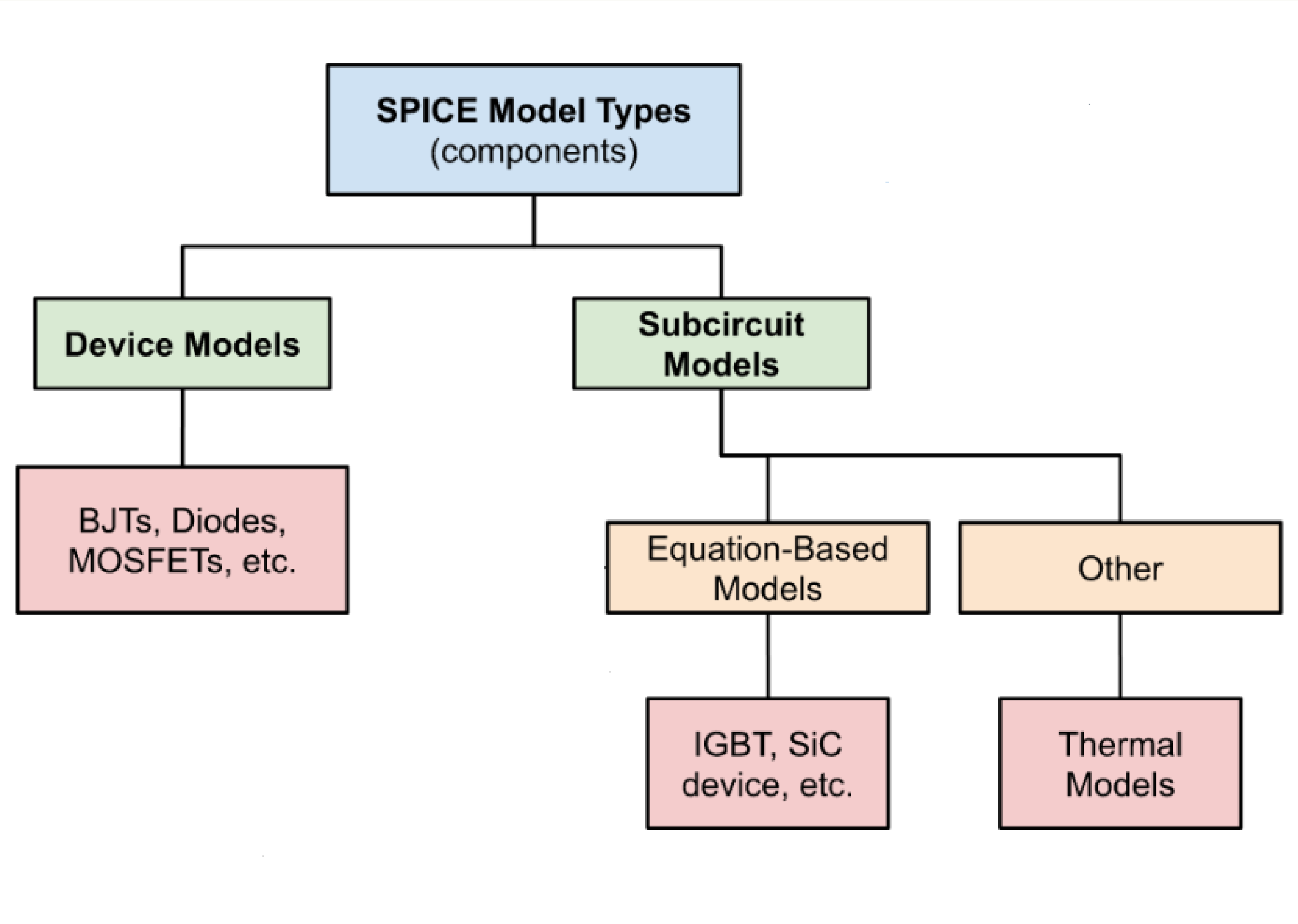
- SPICE was developed at the Electronics Research Laboratory of the University of California, Barkeley by Laurence Nagel in 1973.

- SPICE simulators use a set of mathematical equations to represent a device's behavior, such as its forward current and capacitance for a diode.

- **Example**

- In a diode SPICE model, the current (I_F) is described by an equation that includes the saturation current (I_S) and the diode's voltage (V_d).

Types of SPICE Models



SPICE Device Models

SPICE includes these analyses:

- AC analysis (linear small-signal frequency domain analysis)
- DC analysis (nonlinear quiescent point calculation)
- DC transfer curve analysis (a sequence of nonlinear operating points calculated while sweeping an input voltage or current, or a circuit parameter)
- Noise analysis (a small signal analysis done using an adjoint matrix technique which sums uncorrelated noise currents at a chosen output point)
- Transfer function analysis (a small-signal input/output gain and impedance calculation)
- Transient analysis (time-domain large-signal solution of nonlinear differential algebraic equations)

Thank You